

CombUCB1 for Matching Bandits: Capacity-Constrained and Non-Linear Reward Extensions

Sai Sharan Shaotong Sun Yiqi Li Bole Yi

Abstract

This report studies two extensions of CombUCB1 for online matching problems. The baseline is a stochastic combinatorial semi-bandit algorithm: at each round, the learner selects a feasible set of ground items, observes the individual rewards of the selected items, and uses upper confidence bounds to balance exploration and exploitation. We first reproduce the original CombUCB1 grid-path experiment as a compact implementation check, focusing on the regret-scaling trend reported in the original paper. We then focus on two extensions. Extension 1 introduces capacity-constrained college matching. Each student can be assigned to at most one college, while each college can accept several students up to a capacity limit. We reduce this many-to-one constraint to a one-to-one assignment problem through virtual college slots and solve the optimistic matching problem using a Hungarian oracle. Crucially, virtual slots are only an optimization device; all statistical estimates remain on real student–college edges. Extension 2 changes the reward model from a linear edge-sum objective to a non-linear matching-level objective with a diversity bonus and pairwise interaction terms. The learner still observes semi-bandit edge rewards, but action selection now requires a non-linear oracle. We present the formulation, pseudocode, regret definitions, and the main experimental evidence for both extensions.

1 Introduction

CombUCB1 is a stochastic combinatorial semi-bandit algorithm in which the learner repeatedly chooses a feasible combinatorial action and observes individual rewards for the selected ground items. Its main advantage is modularity: the learning component estimates item-level rewards using upper confidence bounds, while the feasibility structure is handled by an offline optimization oracle [1]. This follows the optimism-under-uncertainty principle from classical stochastic bandits [2] and extends naturally to combinatorial action spaces [3, 4].

This report applies the same principle to matching bandits. Related work has also studied bandit learning in matching markets and decentralized or two-sided settings [5–9]. In our interpretation, each ground item is a student–college edge (s, c) . Selecting the edge means assigning student s to college c and observing the reward of that assignment. A matching is a feasible set of such edges. This perspective is useful because matching constraints are combinatorial, while rewards are learned from repeated interaction.

The project has three parts. First, we reproduce the orig-

inal CombUCB1 behaviour as an implementation check. Second, we extend the feasible set from one-to-one matching to capacity-constrained many-to-one matching, which is natural in college admissions markets [10–13]. Third, we extend the objective from a linear sum of edge rewards to a non-linear matching-level utility, which is connected to set-function optimization where the value of a set may not decompose into independent item values [14]. The first extension changes the constraint structure but keeps a linear oracle. The second extension keeps semi-bandit edge feedback but changes the optimization objective, so the standard linear Hungarian oracle is no longer exact.

2 Baseline CombUCB1 and Reproduction Check

The ground set of arms is denoted by $\mathcal{E} = \{1, \dots, L\}$, and the feasible action family is denoted by $\mathcal{M} \subseteq 2^{\mathcal{E}}$. At round t , the learner chooses an action $A_t \in \mathcal{M}$. Each item $e \in \mathcal{E}$ has an unknown mean reward $\mu_e \in [0, 1]$. In the standard linear semi-bandit model, the reward of an action is

$$r_t(A_t) = \sum_{e \in A_t} X_t(e), \quad (1)$$

where $X_t(e)$ is the stochastic reward of item e at time t . After playing A_t , the learner observes $X_t(e)$ for every selected item $e \in A_t$. This is semi-bandit feedback: the learner observes the rewards of selected components, but not the rewards of unselected components.

If the true means were known, the best fixed feasible action would be

$$A^* \in \arg \max_{A \in \mathcal{M}} \sum_{e \in A} \mu_e. \quad (2)$$

The cumulative regret after T rounds is

$$\text{Reg}(T) = \sum_{t=1}^T \left[\sum_{e \in A^*} \mu_e - \sum_{e \in A_t} \mu_e \right]. \quad (3)$$

CombUCB1 maintains, for each item e , an observation count $N_t(e)$ and empirical mean $\hat{\mu}_t(e)$. After initialization, its optimistic estimate is

$$U_t(e) = \hat{\mu}_t(e) + \sqrt{\frac{1.5 \log t}{N_t(e)}}. \quad (4)$$

The next action is chosen by an oracle:

$$A_t \in \arg \max_{A \in \mathcal{M}} \sum_{e \in A} U_t(e). \quad (5)$$

Thus, the algorithm does not need to know how to solve every combinatorial problem internally; it only needs an oracle for the deterministic problem with weights $U_t(e)$.

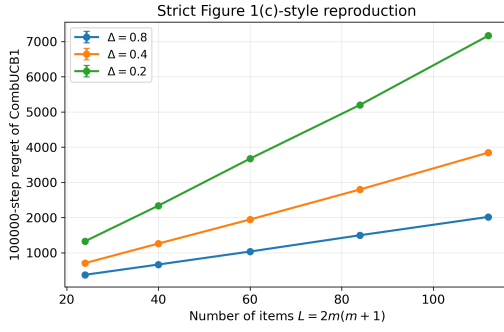


Figure 1: Reproduction check for the original CombUCB1 Figure 1(c)-style grid-path experiment. The plot is used to validate the expected regret-scaling trend before introducing the matching extensions.

Figure 1 is the only reproduction figure kept in the main text. It targets the original grid-path experiment rather than a generic cumulative-regret sanity plot. In that experiment, the ground items are the directed grid edges, the feasible actions are source-to-target paths, and the optimal path is made slightly better than the alternatives through a reward gap parameter. The original empirical question is whether regret scales in the expected direction when the number of items grows and when the gap becomes smaller. Our reproduction therefore focuses on the same qualitative scaling behaviour: larger instances and smaller gaps should produce larger cumulative regret.

The reproduction is designed as an implementation-level validation of the original experimental mechanism, rather than as an exact visual duplication of the published figure. For a fixed horizon and repeated synthetic trials, the expected qualitative behaviour is that cumulative regret increases when the grid contains more items and when the reward gap becomes harder to distinguish. This level of reproduction is appropriate for the report because the main contribution is not a new proof of the original regret bound, but the two matching extensions built on top of the same learning loop.

For the reproduction part, we follow the original CombUCB1 covering initialization idea: before the main UCB loop, every feasible ground item is observed at least once through oracle calls that prioritize currently unobserved items. In the extension pseudocode below, this is implemented by assigning infinite UCB to unobserved real edges together with deterministic tie-breaking that prioritizes uncovered edges. This avoids division by zero and prevents the implementation from repeatedly selecting only a small subset of initially unseen arms.

The reproduction experiment also fixes the interface used by the extensions. In all three parts of the report, the learner maintains item-level statistics, converts these statistics into optimistic weights, calls an optimization oracle, observes semi-bandit feedback on the selected items, and updates only the selected item statistics. Thus, the extensions below do not change the basic CombUCB1 learning loop. They change either the feasible family handled by the oracle or the objective optimized by the oracle. This separation is impor-

tant because it keeps the learning rule and the optimization model mathematically consistent.

Implementation fidelity. The reproduction was implemented as a controlled validation of the algorithmic interface rather than as an attempt to rederive the original proof. The empirical mean and count of each ground item are updated only after that item is observed, and the oracle receives only the current optimistic weights. This convention is preserved throughout the extensions. In particular, additional modelling structure is introduced at the oracle level, not by altering the statistical update rule. This design choice is important because it makes the comparison across the baseline, the capacity-constrained model, and the non-linear model interpretable.

3 Extension 1: Capacity-Constrained Matching

3.1 Problem formulation

The student set is denoted by $\mathcal{S}$, and the college set is denoted by $\mathcal{C}$. Each real edge $(s, c) \in \mathcal{S} \times \mathcal{C}$ is a ground arm with unknown mean reward $\mu_{s,c}$. A feasible capacity-constrained matching M must satisfy

$$\sum_{c \in \mathcal{C}} \mathbf{1}\{(s, c) \in M\} \leq 1, \quad \forall s \in \mathcal{S}, \quad (6)$$

$$\sum_{s \in \mathcal{S}} \mathbf{1}\{(s, c) \in M\} \leq k_c, \quad \forall c \in \mathcal{C}. \quad (7)$$

Equation (6) means that each student can be assigned to at most one college. Equation (7) means that college c can accept up to k_c students. This is the many-to-one structure missing from a simple one-to-one matching formulation.

Under a linear reward model, the value of a matching is

$$\mu(M) = \sum_{(s,c) \in M} \mu_{s,c}. \quad (8)$$

The learning problem remains semi-bandit: after selecting M_t , the learner observes the individual edge rewards $X_t(s, c)$ for $(s, c) \in M_t$.

3.2 Virtual slots and the Hungarian oracle

The central reduction replaces every college c with k_c identical virtual slots:

$$\mathcal{V}_c = \{v_{c,1}, \dots, v_{c,k_c}\}, \quad \mathcal{V} = \bigcup_{c \in \mathcal{C}} \mathcal{V}_c. \quad (9)$$

A student-slot edge $(s, v_{c,j})$ represents assigning student s to one seat of college c . Its optimistic weight is copied from the real edge:

$$W_t(s, v_{c,j}) = U_t(s, c). \quad (10)$$

The Hungarian algorithm solves a one-to-one assignment problem over students and slots [15, 16]. After the assignment is found, each selected slot edge $(s, v_{c,j})$ is projected back to the real edge (s, c) .

The key modelling requirement is that virtual slots must not become separate statistical arms. The learner maintains one count and one empirical mean per real edge:

$$N_t(s, c), \quad \hat{\mu}_t(s, c), \quad (11)$$

not separate statistics $N_t(s, c, j)$ for different slots. Otherwise, the same real assignment would be artificially split into several arms, duplicating exploration and changing the meaning of the bandit problem.

If the model allows some students to remain unmatched, dummy slots with zero weight are added before running the assignment algorithm. These dummy assignments are discarded afterwards. This prevents the Hungarian formulation from forcing every student into a real college slot when the original matching constraint only requires “at most one” assignment.

The virtual-slot construction changes only the deterministic optimization problem solved by the oracle. It does not change the statistical learning problem. For example, if college c has three available slots, the oracle may contain three slot edges $(s, v_{c,1})$, $(s, v_{c,2})$, and $(s, v_{c,3})$. However, all three correspond to the same real assignment (s, c) and therefore share the same reward distribution, empirical mean, and observation count. Consequently, the update step below projects the selected slot edge back to a real edge before updating $N(s, c)$ and $S(s, c)$.

This construction also clarifies the modelling boundary of the extension. The capacity constraint changes which sets of edges are feasible, but the stochastic object associated with an edge remains unchanged. The reduction is therefore an oracle-level reduction from a many-to-one assignment to a one-to-one assignment over virtual seats. It should not be interpreted as introducing multiple independent copies of the same college assignment.

Algorithm 1 makes the separation explicit. Lines 4–11 compute UCB values on real edges. Lines 13–16 solve a deterministic assignment problem on the expanded slot graph. Lines 17–21 return to the real graph and update only the arms that were actually observed. Therefore, the Hungarian algorithm is used as an oracle for capacity feasibility, not as a replacement for the bandit learning rule.

3.3 Computational cost of the capacity oracle

Denote the number of students by $n_S = |S|$, the number of colleges by $n_C = |C|$, and the total number of virtual college slots by $K_C = \sum_{c \in C} k_c$. Computing UCB values for all real student–college edges costs $O(n_S n_C)$ per round. Constructing the student–slot weight matrix costs $O(n_S K_C)$ because every real college edge is copied to each of its virtual slots. The dominant step is the assignment oracle. A standard Hungarian implementation on the rectangular student–slot matrix has polynomial cost, commonly written as $O(\max\{n_S, K_C\}^3)$ after padding to a square matrix. Therefore, the per-round cost of Extension 1 is

$$O(n_S n_C + n_S K_C + \max\{n_S, K_C\}^3). \quad (12)$$

This cost analysis also clarifies the role of virtual slots: they increase the size of the deterministic oracle problem, but they do not increase the number of statistical arms beyond the real edges (s, c) .

3.4 Regret and interpretation

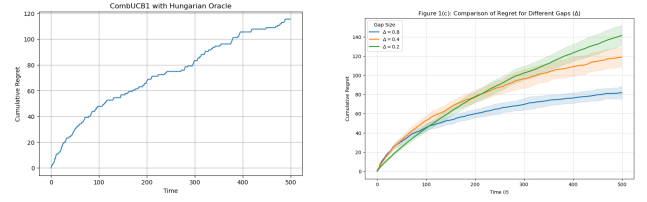
The optimal capacity-feasible matching under the true means is

$$M_{\text{cap}}^* \in \arg \max_{M \in \mathcal{M}_{\text{cap}}} \sum_{(s,c) \in M} \mu_{s,c}, \quad (13)$$

where $\mathcal{M}_{\text{cap}}$ denotes the family of matchings satisfying Equations (6) and (7). The regret is

$$\text{Reg}_{\text{cap}}(T) = \sum_{t=1}^T \left[\sum_{(s,c) \in M_{\text{cap}}^*} \mu_{s,c} - \sum_{(s,c) \in M_t} \mu_{s,c} \right]. \quad (14)$$

This is still a linear semi-bandit regret, but over a richer feasible family.



(a) Capacity-constrained Hungarian oracle.

(b) Gap-dependent regret.

Figure 2: Extension 1 results. The left plot validates that CombUCB1 can be combined with a Hungarian oracle after converting college capacities into virtual slots. The right plot shows the expected gap-dependent behaviour: smaller reward gaps make it harder to identify the optimal matching and therefore lead to larger regret.

Figure 2 contains the two figures needed for the Extension 1 argument. The first shows that the capacity-constrained oracle works inside CombUCB1. The second is important because it tests the central UCB intuition: regret should increase when the optimal and near-optimal matchings become harder to distinguish.

4 Extension 2: Non-Linear Matching Rewards

4.1 Objective

The baseline and Extension 1 both assume that the value of a matching is additive over selected edges. This is restrictive. In many matching problems, the quality of a set of assignments may depend on the composition of the selected set. For example, a matching may receive a diversity bonus, or two assignments may interact positively or negatively.

We model this by defining the value of a matching as

$$F(M; \mu) = \sum_{(s,c) \in M} \mu_{s,c} + \lambda \text{Div}(M) + \gamma \text{Int}(M). \quad (15)$$

The first term is the learned edge-reward component. The diversity term $\text{Div}(M)$ is a known deterministic function of the selected matching. In our implementation, colleges are assigned to fixed categories and $\text{Div}(M)$ is computed as a concave coverage bonus over the category counts, so selecting colleges from several categories can increase the value of the whole matching. The interaction term $\text{Int}(M)$

Algorithm 1 Extension 1: CombUCB1 with virtual slots and a Hungarian oracle

Require: Students $\mathcal{S}$, colleges $\mathcal{C}$, capacities k_c , horizon T .

```
1: Initialize real-edge statistics  $N(s, c) \leftarrow 0$  and  $S(s, c) \leftarrow 0$  for every  $(s, c)$ .
2: Run a covering initialization: prioritize unobserved real edges until every feasible real edge has been sampled at least once.
3: for  $t = 1, 2, \dots, T$  do
4:   for each real edge  $(s, c)$  do
5:     if  $N(s, c) = 0$  then
6:        $U(s, c) \leftarrow +\infty$ .
7:     else
8:        $\hat{\mu}(s, c) \leftarrow S(s, c)/N(s, c)$ .
9:        $U(s, c) \leftarrow \hat{\mu}(s, c) + \sqrt{1.5 \log(t)/N(s, c)}$ .
10:    end if
11:  end for
12:  Replace each college  $c$  by virtual slots  $v_{c,1}, \dots, v_{c,k_c}$ .
13:  Assign slot weights  $W(s, v_{c,j}) \leftarrow U(s, c)$  for all  $j$ .
14:  Add zero-weight dummy slots if unmatched students are allowed.
15:  Run the Hungarian algorithm on the student-slot weight matrix.
16:  Project selected slot edges  $(s, v_{c,j})$  back to real edges  $(s, c)$  to obtain  $M_t$ .
17:  Play  $M_t$  and observe  $X_t(s, c)$  for each selected real edge.
18:  for each observed real edge  $(s, c) \in M_t$  do
19:     $N(s, c) \leftarrow N(s, c) + 1$ .
20:     $S(s, c) \leftarrow S(s, c) + X_t(s, c)$ .
21:  end for
22: end for
```

is computed from fixed pairwise interaction scores between the colleges selected in the matching. Both terms are fixed before learning starts and are not estimated from bandit feedback. The parameters λ and γ control how strongly these matching-level terms affect the objective. A concrete version used in the experiments is

$$\text{Div}(M) = \sum_a \sqrt{n_a(M)}, \quad (16)$$

$$\text{Int}(M) = \frac{1}{\max\{1, |M|\}} \sum_{\substack{(s,c),(s',c') \in M \\ (s,c) \prec (s',c')}} B_{c,c'}. \quad (17)$$

Here $n_a(M)$ is the number of selected colleges from category a , $B_{c,c'}$ is a fixed pairwise college interaction score, $B_{c,c} = 0$ when two selected assignments involve the same college, and $\prec$ is any fixed ordering used only to count each unordered pair once. The normalization by $\max\{1, |M|\}$ keeps the interaction term on a comparable scale when the matching size changes. The exact numerical values of categories and pairwise scores are part of the synthetic instance, not random feedback observed online.

Only the edge means are stochastic and unknown. Therefore, the UCB bonus is applied only to the edge-reward term:

$$F_t^{\text{UCB}}(M) = \sum_{(s,c) \in M} U_t(s, c) + \lambda \text{Div}(M) + \gamma \text{Int}(M). \quad (18)$$

This distinction is important. The learner does not estimate a noisy diversity reward or a noisy interaction reward. It learns only the unknown edge means from semi-bandit feedback and uses the known matching-level terms during action selection and evaluation. Therefore, the extension is not claiming to learn an arbitrary unknown non-linear reward function. Its purpose is to show how CombUCB1 changes when the learned edge means enter a non-additive matching-level utility.

This formulation is deliberately conservative. If the diversity or interaction terms were also unknown and stochastic, the feedback model would need additional observations or a separate estimator for those terms. Such a model would no longer be a direct extension of the original semi-bandit setting. By keeping the matching-level terms fixed and known, the experiment isolates a single issue: the offline oracle must optimize a non-additive objective even though the statistical uncertainty remains edge-level.

4.2 Why the Hungarian oracle is no longer exact

The Hungarian algorithm solves additive assignment objectives of the form

$$\max_M \sum_{(s,c) \in M} w_{s,c}. \quad (19)$$

Equation (18) is not generally additive because $\text{Div}(M)$ and $\text{Int}(M)$ depend on the whole matching. Therefore, one cannot simply put these terms into independent edge weights without changing the problem. For this reason, Extension 2 uses an exact enumeration oracle on small synthetic instances:

$$M_t \in \arg \max_{M \in \mathcal{M}} F_t^{\text{UCB}}(M). \quad (20)$$

The enumeration oracle is not intended as a scalable production method. It is used to isolate the modelling effect of changing the reward objective from linear to non-linear. For larger instances, this oracle would need to be replaced by an approximate non-linear optimization method.

The essential point is that the non-linear terms are matching-level terms. A linear oracle can rank individual edges independently and then combine them, but here the marginal contribution of one edge may depend on which other edges are already selected. For example, adding a college from a new category can increase the diversity bonus

more than adding another college from an already represented category. Similarly, the interaction contribution of choosing college c depends on the other selected colleges through $B_{c,c'}$. This dependency is exactly what breaks the additive assignment structure in Equation (19).

Algorithm 2 Extension 2: CombUCB1 with a non-linear matching oracle

Require: Feasible matchings $\mathcal{M}$, horizon T , parameters λ, γ .

- 1: Initialize $N(s, c) \leftarrow 0$ and $S(s, c) \leftarrow 0$ for every edge.
- 2: Enumerate all feasible matchings $\mathcal{A} = \{M_1, \dots, M_R\}$ for the small synthetic instance.
- 3: Run the covering initialization so each feasible edge is observed at least once.
- 4: **for** $t = 1, 2, \dots, T$ **do**
- 5: **for** each edge (s, c) **do**
- 6: **if** $N(s, c) = 0$ **then**
- 7: $U(s, c) \leftarrow +\infty$.
- 8: **else**
- 9: $\hat{\mu}(s, c) \leftarrow S(s, c)/N(s, c)$.
- 10: $U(s, c) \leftarrow \hat{\mu}(s, c) + \sqrt{1.5 \log(t)/N(s, c)}$.
- 11: **end if**
- 12: **end for**
- 13: **for** each matching $M \in \mathcal{A}$ **do**
- 14: Score M by $F_t^{\text{UCB}}(M)$ from Eq. (18).
- 15: **end for**
- 16: Select $M_t \in \arg \max_{M \in \mathcal{A}} F_t^{\text{UCB}}(M)$.
- 17: Play M_t and observe $X_t(s, c)$ for all $(s, c) \in M_t$.
- 18: **for** each observed edge $(s, c) \in M_t$ **do**
- 19: $N(s, c) \leftarrow N(s, c) + 1$; $S(s, c) \leftarrow S(s, c) + X_t(s, c)$.
- 20: **end for**
- 21: **end for**

Algorithm 2 differs from Algorithm 1 only in the oracle step. The update rule is still edge-level semi-bandit learning: after a matching is played, only the observed edge rewards update $N(s, c)$ and $S(s, c)$. The diversity and interaction terms enter selection through $F_t^{\text{UCB}}(M)$, but they do not create additional stochastic arms. This distinction prevents the modelling error in which the non-linear bonus should affect which matching is chosen, but it should not be treated as an additional noisy edge reward.

4.3 Regret and evaluation

The optimal matching under the true non-linear objective is

$$M_{\text{nl}}^* \in \arg \max_{M \in \mathcal{M}} F(M; \mu). \quad (21)$$

The cumulative regret is

$$\text{Reg}_{\text{nl}}(T) = \sum_{t=1}^T [F(M_{\text{nl}}^*; \mu) - F(M_t; \mu)]. \quad (22)$$

When comparing the non-linear oracle with a linear CombUCB1 baseline, both methods must be evaluated using Equation (22). The linear baseline may choose actions using only the edge-sum UCB objective, but its performance is still measured under the true non-linear objective. Otherwise, the comparison would use different objectives for different algorithms and would not be meaningful.

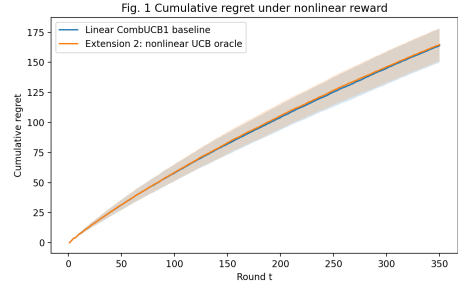


Figure 3: Extension 2 main result. Both methods learn edge-level rewards from semi-bandit feedback and are evaluated under the same true non-linear objective. The comparison isolates the effect of replacing the linear oracle with a non-linear matching oracle.

Figure 3 is the main figure for Extension 2. Sensitivity, ablation, and runtime plots may be useful for implementation diagnostics, but the central claim is already captured by this comparison: once the reward is no longer additive over edges, the action-selection oracle must change.

5 Experimental Protocol and Interpretation

All experiments use synthetic instances because regret requires access to the true reward means. In a real admissions or recommendation dataset, the learner would not know the reward of every unselected edge, so exact regret could not be computed without additional assumptions. Synthetic data allow us to compute the offline optimum and measure cumulative regret directly.

Table 1: Compact experimental protocol.

Experiment	Oracle / metric	Claim checked
Reproduction	Path oracle; linear regret	Baseline scaling trend.
Capacity matching	Hungarian oracle; linear regret	Capacities without duplicated arms.
Gap test	Hungarian oracle; regret across gaps	Smaller gaps increase difficulty.
Non-linear matching	Enumeration oracle; non-linear regret	Non-additive utility needs a non-linear oracle.

The reproduction experiment validates the baseline implementation before applying the two extensions. The selected reproduction plot is the Figure 1(c)-style scaling result rather than a generic time-series plot, because it is closer to the original experiment and therefore stronger evidence that the pipeline is aligned with the original CombUCB1 setting.

For Extension 1, the main experimental question is whether the many-to-one capacity constraint can be handled without changing the statistical meaning of the arms. The virtual-slot transformation makes the oracle efficient, while real-edge statistics preserve the correct learning problem. The gap-dependent result is included because it checks whether the UCB mechanism behaves as expected when the difficulty of the instance changes.

For Extension 2, the main experimental question is whether a non-linear matching objective changes the behaviour of the learner. The exact enumeration oracle is used only on small synthetic instances. This keeps the experiment

exact and interpretable, but it also exposes the scalability limitation of the extension. A large-scale version would need a heuristic or approximate non-linear oracle.

The reported plots are selected to support the modelling claims rather than to exhaustively report every diagnostic generated during development. The reproduction figure checks the baseline implementation against the original experiment. The two Extension 1 plots check the capacity oracle and the expected gap-dependent UCB behaviour. The Extension 2 plot checks whether using the non-linear oracle changes regret under the same true non-linear objective. Additional sensitivity and runtime figures are useful for implementation diagnostics, but they are secondary to the central modelling contributions.

Reproducibility of the synthetic setup. The synthetic design has two advantages for this report. First, the true mean vector is available, so the offline optimal matching can be computed exactly for the regret calculation. Second, the difficulty of the instance can be controlled directly by changing the reward gaps, capacities, or the strengths of the diversity and interaction terms. This makes the experiments suitable for validating modelling assumptions, although it also means that the results should be interpreted as controlled algorithmic evidence rather than as claims about a particular real admissions market.

How the regret curves should be read. The regret curves are not intended to establish a new theoretical regret theorem. Their role is to check whether the implementation is consistent with the modelling claim of each section. In the reproduction experiment, the relevant observation is the direction of the scaling trend, not an exact match of every numerical value in the original figure. In Extension 1, the important observation is that the same edge-level UCB updates remain valid after the capacity constraint is pushed into the oracle. In Extension 2, the important observation is that an algorithm selecting actions under the non-linear objective should be evaluated under that same objective; otherwise, the comparison would reward the wrong behaviour.

Relation to the original CombUCB1 theory. The original regret bounds apply to stochastic combinatorial semi-bandits with a linear reward over selected items and an exact offline oracle. Extension 1 still fits this template after the virtual-slot reduction, because the reward remains a sum of real-edge means and the Hungarian algorithm gives an exact solution to the expanded assignment problem. The only additional care is that the statistical arms are the real edges, not the duplicated slot edges. Extension 2 deliberately moves outside the purely linear template: the learner still observes semi-bandit edge feedback, but the action value contains known matching-level terms. Therefore, the original theorem should not be quoted as a complete proof for Extension 2. Instead, the experiment is presented as an exact small-instance study of how the oracle must change when the objective is non-additive.

6 Discussion and Limitations

Extension 1 is a structural extension. It changes the feasible family from one-to-one matching to many-to-one matching with college capacities. The key idea is the virtual-slot reduction. The main risk is statistical: if virtual slots are treated as distinct arms, the same real assignment is duplicated and learning becomes incorrectly defined. Our formulation avoids this by keeping all statistics on real student-college edges.

Extension 2 is an objective-level extension. It keeps semi-bandit feedback on edges but changes the value of a matching by adding known matching-level terms. This makes the standard linear oracle insufficient. The exact enumeration oracle is correct for small instances, but it does not scale. This limitation is a direct consequence of moving from an additive objective to a genuinely non-linear objective, rather than an implementation defect. A larger implementation would need to replace enumeration with a greedy, local-search, mixed-integer, or other approximate non-linear oracle. In that case, the clean regret comparison used here would also need to specify whether regret is measured against the exact optimum or against the best solution reachable by the chosen approximation oracle.

A second limitation is that all experiments are synthetic. This is intentional because cumulative regret requires the true means of both selected and unselected edges. In a real matching market, unobserved assignments would not reveal their counterfactual rewards. Real data would therefore be useful for qualitative evaluation, but exact regret curves would require either a simulator, logged bandit feedback assumptions, or an offline evaluation protocol.

The two extensions therefore demonstrate different ways of adapting CombUCB1. Extension 1 enriches the feasible set while preserving a linear oracle. Extension 2 enriches the reward model but requires a more expressive oracle. Taken together, the two extensions show that the modular learning-plus-oracle view of CombUCB1 is useful, but the oracle must match the structure of the chosen action space and reward objective.

Internal consistency of the algorithms. The formulas and pseudocode use the same three invariants throughout the paper. First, counts and empirical sums are always indexed by real edges, never by virtual slots or by whole matchings. Second, UCB bonuses are applied only to stochastic quantities that are actually observed through semi-bandit feedback. Third, every regret definition evaluates the algorithm under the same objective that defines the corresponding offline optimum. These invariants are what make the capacity extension and the non-linear extension mathematically consistent with the baseline CombUCB1 loop.

7 Conclusion

This report reproduces the original CombUCB1 behaviour as an implementation check and then develops two matching-bandit extensions. Extension 1 introduces capacity-constrained college matching. The virtual-slot transformation allows the optimistic matching problem to be solved

by a Hungarian oracle, while all learning statistics remain correctly defined on real student–college edges. Extension 2 introduces a non-linear matching-level reward with diversity and interaction terms. Because this objective is not additive over edges, the standard Hungarian oracle is no longer exact; the implementation therefore uses exact enumeration for small synthetic instances. These two extensions highlight the same central lesson: CombUCB1 can be adapted to richer matching problems, but the feasible set, reward definition, oracle, and regret metric must remain mathematically consistent.

References

- [1] Branislav Kveton, Zheng Wen, Azin Ashkan, and Csaba Szepesvari. Tight regret bounds for stochastic combinatorial semi-bandits. In *Proceedings of AIS-TATS*, pages 535–543, 2015.
- [2] Peter Auer, Nicolò Cesa-Bianchi, and Paul Fischer. Finite-time analysis of the multiarmed bandit problem. *Machine Learning*, 47(2):235–256, 2002.
- [3] Yi Gai, Bhaskar Krishnamachari, and Rahul Jain. Combinatorial network optimization with unknown variables. *IEEE/ACM Transactions on Networking*, 20(5):1466–1478, 2012.
- [4] Wei Chen, Yajun Wang, and Yang Yuan. Combinatorial multi-armed bandit: General framework and applications. In *Proceedings of ICML*, pages 151–159, 2013.
- [5] Lydia Liu, Horia Mania, and Michael Jordan. Competing bandits in matching markets. In *Proceedings of AISTATS*, pages 1618–1628, 2020.
- [6] Lydia Liu, Feng Ruan, Horia Mania, and Michael Jordan. Bandit learning in decentralized matching markets. *Journal of Machine Learning Research*, 22(211):1–34, 2021.
- [7] Z. Wang, L. Guo, J. Yin, and S. Li. Bandit learning in many-to-one matching markets. In *Proceedings of CIKM*, pages 2088–2097, 2022.
- [8] Y. Zhang and Z. Fang. Decentralized competing bandits in many-to-one matching markets. In *Proceedings of AAMAS*, pages 2603–2605, 2024.
- [9] Y. Li, G. Cheng, and X. Dai. Two-sided competing matching recommendation markets. *arXiv preprint arXiv:2301.10230*, 2023.
- [10] Alvin E. Roth. The college admissions problem is not equivalent to the marriage problem. *Journal of Economic Theory*, 36(2):277–288, 1985.
- [11] Alvin E. Roth and Marilda Sotomayor. The college admissions problem revisited. *Econometrica*, 57(3):559–570, 1989.
- [12] Alvin E. Roth and Marilda Sotomayor. Two-sided matching. In *Handbook of Game Theory with Economic Applications*, volume 1, pages 485–541. Elsevier, 1992.
- [13] Tayfun Sönmez. Strategy-proofness in many-to-one matching problems. *Economic Design*, 1:365–380, 1994.
- [14] George Nemhauser, Laurence Wolsey, and Marshall Fisher. An analysis of approximations for maximizing submodular set functions. *Mathematical Programming*, 14:265–294, 1978.
- [15] Harold W. Kuhn. The hungarian method for the assignment problem. *Naval Research Logistics Quarterly*, 2:83–97, 1955.
- [16] James Munkres. Algorithms for the assignment and transportation problems. *Journal of the Society for Industrial and Applied Mathematics*, 5(1):32–38, 1957.